

**10.1** Let's combine the two terms:

$$\begin{aligned}
& \mathcal{L}(q) + KL(q||p) \\
&= \left( \int q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z})}{q(\mathbf{Z})} \right\} d\mathbf{Z} \right) + \left( - \int q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{Z}|\mathbf{X})}{q(\mathbf{Z})} \right\} d\mathbf{Z} \right) \\
&= \int q(\mathbf{Z}) \left( \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z})}{q(\mathbf{Z})} \right\} - \ln \left\{ \frac{p(\mathbf{Z}|\mathbf{X})}{q(\mathbf{Z})} \right\} \right) d\mathbf{Z} \\
&= \int q(\mathbf{Z}) \left( \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z})}{q(\mathbf{Z})} \cdot \frac{q(\mathbf{Z})}{p(\mathbf{Z}|\mathbf{X})} \right\} \right) d\mathbf{Z} \\
&= \int q(\mathbf{Z}) \left( \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z})}{p(\mathbf{Z}|\mathbf{X})} \right\} \right) d\mathbf{Z} \\
&= \int q(\mathbf{Z}) \ln p(\mathbf{X}) d\mathbf{Z} \\
&= \ln p(\mathbf{X}) \int q(\mathbf{Z}) d\mathbf{Z} \\
&= \ln p(\mathbf{X})
\end{aligned}$$